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Anti-lock braking system for utility vehicle

Abstract

A utility vehicle includes a frame and a plurality of ground-engaging members supporting the frame. Each of the plurality of ground-engaging members is configured to rotate about an axle. The utility vehicle further includes a powertrain assembly supported by the frame and a braking system configured to operate in a normal run mode and an anti-lock braking mode. The braking system includes an anti-lock braking control module operably coupled to the plurality of ground-engaging members and configured to automatically engage the anti-lock braking mode in response to a predetermined condition.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/193,504, filed Mar. 5, 2021, which is a continuation of U.S. patent application Ser. No. 16/197,497, filed Nov. 21, 2018, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/590,041, filed Nov. 22, 2017, and entitled “ANTI-LOCK BRAKING SYSTEM FOR UTILITY VEHICLE,” the entirety of which are hereby incorporated herein by reference.

TECHNICAL FIELD OF THE DISCLOSURE
(1) The present application relates to a braking system for a vehicle and, more particularly, to an anti-lock braking system for a utility vehicle configured for off-road applications.
BACKGROUND OF THE DISCLOSURE
(2) Anti-lock braking systems (“ABS”) may be used on vehicles to facilitate braking power in response to a user input. For example, the user may depress a brake pedal, thereby enabling the ABS to facilitate braking of the vehicle. The ABS may be configured to facilitate braking of the front wheels, the rear wheels, or both the front and rear wheels.
(3) In some embodiments, it may be possible to disengage the ABS. However, if a user disengages or turns off the ABS, then the user may have to remember to manually re-engage or turn on the ABS when needed. In such instances, the user must be sufficiently cognizant of the terrain, driving, and other conditions to recognize that the ABS should be turned on before it is needed. As such, there is a need for a system which may automatically engage or turn on the ABS in response to a predetermined driving condition.
SUMMARY OF THE DISCLOSURE
(4) In one embodiment, a utility vehicle comprises a frame and a plurality of ground-engaging members supporting the frame. Each of the plurality of ground-engaging members is configured to rotate about an axle. The utility vehicle further comprises a powertrain assembly supported by the frame and a braking system configured to operate in a normal run mode and an anti-lock braking mode. The braking system comprises an anti-lock braking control module operably coupled to the plurality of ground-engaging members and configured to automatically engage the anti-lock

braking mode in response to a predetermined condition.
(5) In another embodiment, a braking assembly for a utility vehicle configured to operate in a normal run mode and an anti-lock braking mode is disclosed. The braking assembly comprises a user braking member, a plurality of brake calipers operably coupled to the user braking member, a junction member operably coupled to at least two of the plurality of brake calipers, and an anti-lock braking control module operably coupled to at least the user braking member and junction member. The anti-lock braking control module is configured to automatically engage the anti-lock braking mode at a predetermined condition and disengage the anti-lock braking mode in response to a user input.
(6) In yet another embodiment, a method of operating a braking assembly of a utility vehicle in one of a normal run mode and an anti-lock braking mode comprises providing a user braking member, providing a plurality of brake calipers operably coupled to the user braking member, providing an anti-lock braking control module operably coupled to the user braking member and the plurality of brake calipers, and automatically engaging the anti-lock braking mode at a predetermined condition.
(7) Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiment exemplifying the best mode of carrying out the invention as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The foregoing aspects and many of the intended advantages of this invention will become more readily appreciated as the same becomes better understood by reference to the following detailed description when taken in conjunction with the accompanying drawings.
(2) FIG. **1** is a left front perspective view of a utility vehicle of the present disclosure;
(3) FIG. **2** is a left rear perspective view of a braking assembly of the utility vehicle of FIG. **1**;
(4) FIG. **3** is a rear perspective view of the braking assembly of FIG. **2**;
(5) FIG. **4** is a right front perspective view of a front portion of the braking assembly of FIG. **2**;
(6) FIG. **5** is a junction member of the braking assembly of FIG. **2**;
(7) FIG. **6** is a left rear perspective view of a front drive member of the utility vehicle of FIG. **1**;
(8) FIG. **7** is a left rear perspective view of a rear drive member of the utility vehicle of FIG. **1**;
(9) FIG. **8** is a schematic view of a portion of an electrical system of the utility vehicle of FIG. **1**;
(10) FIG. **9** is a schematic view of an electronic braking circuit of the electrical system of FIG. **8**;
(11) FIG. **10**A is a schematic view of a hydraulic circuit of the braking assembly of FIG. **2**;
(12) FIG. **10**B is a schematic view of an alternative hydraulic circuit of the braking assembly of FIG. **2**;
(13) FIG. **11** is a control diagram of the braking assembly of FIG. **2** operating in a first or ABS On operating mode;
(14) FIG. **12** is a control diagram of the braking assembly of FIG. **2** operating in a second or ABS On-Off operating mode;
(15) FIG. **13** is a control diagram of the braking assembly of FIG. **2** operating in a third or ABS Control operating mode;
(16) FIGS. **14**A and **14**B are respective portions of a control diagram of an adjustable speed limiting feature of the vehicle of FIG. **1**.
(17) FIG. **15** is a control diagram of an electronic stability control (“ESC”) assembly of the vehicle of FIG. **1** operating in a first or Normal ESC and ABS Mode;
(18) FIG. **16** is a control diagram of the ESC assembly operating in a second or Hill Descent Control (“HDC”) Mode;

(19) FIG. **17** is a control diagram of the ESC assembly operating in a third or Hill Assist/Hill Hold Control ("HHC") Mode;
(20) FIG. **18** is a control diagram of the ESC assembly operating in a fourth or Roll Over Mitigation ("ROM") Mode;
(21) FIG. **19** is a control diagram of the ESC assembly operating in a fifth or Traction Control System ("TCS") Mode; and
(22) FIG. **20** is a control diagram of the ESC assembly operating in a sixth or Vehicle Dynamic Control ("VDC") Mode.
(23) Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components according to the present disclosure, the drawings are not necessarily to scale and certain features may be exaggerated in order to better illustrate and explain the present disclosure. The exemplifications set out herein illustrate embodiments of the invention, and such exemplifications are not to be construed as limiting the scope of the invention in any manner.
DETAILED DESCRIPTION OF THE DRAWINGS
(24) For the purposes of promoting an understanding of the principals of the invention, reference will now be made to the embodiments illustrated in the drawings, which are described below. The embodiments disclosed below are not intended to be exhaustive or limit the invention to the precise form disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings. It will be understood that no limitation of the scope of the invention is thereby intended. The invention includes any alterations and further modifications in the illustrative devices and described methods and further applications of the principles of the invention which would normally occur to one skilled in the art to which the invention relates.
(25) As shown in FIG. **1**, a utility vehicle **2** is disclosed and configured for off-road vehicle applications, such that utility vehicle **2** is configured to traverse trails and other off-road terrain. Utility vehicle **2** includes a frame assembly **4** which supports a plurality of body panels **6** and is supported on a ground surface by a plurality of ground-engaging members **8**. Illustratively, ground-engaging members **8** include front ground-engaging members **10** and rear ground-engaging members **12**. In one embodiment of vehicle **2**, each of front ground-engaging members **10** include a wheel assembly **10***a* and a tire **10***b* supported thereon. Similarly, each of rear ground-engaging members **12** may include a wheel assembly **12***a* and a tire **12***b* supported thereon. A front suspension assembly **27** may be operably coupled to front ground-engaging members **10** and a rear suspension assembly **28** may be operably coupled to rear ground-engaging members **12**.
(26) Referring still to FIG. **1**, utility vehicle **2** extends between a front end portion **14** and a rear end portion **16** along a longitudinal axis L and supports an operator area **18** therebetween. Operator area **18** includes seating **20** for at least the operator and also may support one or more passengers. In one embodiment, seating **20** includes side-by-side bucket-type seats while, in another embodiment, seating **20** includes a bench-type seat. A cargo area **22** is positioned rearward of operator area **18** and is supported by frame assembly **4** at rear end portion **16**.
(27) As shown in FIG. **1**, operator area **18** includes operator controls **24**, such as steering assembly **26**, which may be operably coupled to one or more of ground-engaging members **8**. Additional operator controls **24** may include other inputs for controlling operation of vehicle **2**, as disclosed further herein, such as an accelerator member or pedal **53** and a brake member or pedal **54** (FIG. **2**). More particularly, various operator controls **24** may affect operation of a powertrain assembly **30** of vehicle **2**. Powertrain assembly **30** may be supported by rear end portion **16** of vehicle **2** and includes an engine (not shown), a transmission (not shown) operably coupled to the engine, a front final drive member **32** (FIG. **2**) operably coupled to front ground-engaging members **10** through front half shafts or axles **37**, and a rear final drive member **34** (FIG. **2**) operably coupled to rear ground-engaging members **12** through rear half shafts or axles **38**. A drive shaft (not shown) may

be operably coupled to front final drive member **32** at an input **36** (FIG. **2**) for supplying motive power from the engine and/or transmission to front ground-engaging members **10**. Rear final drive member **34** is operably coupled the engine and/or transmission to supply power therefrom to rear ground-engaging members **12**.

(28) Referring to FIGS. **2-4**, vehicle **2** includes a braking assembly **40**, illustratively an anti-lock braking system ("ABS"), which includes a front end braking portion **42** positioned generally at front end portion **14** of vehicle **2** and is operably coupled to front ground-engaging members **10** and a rear end braking portion **44** positioned generally at rear end portion **16** of vehicle **2** and is operably coupled to rear ground-engaging members **12**. Front end braking portion **42** includes front brake discs **46** and front brake calipers **48** operably coupled to front wheel assemblies **10***a*. Rear end braking portion **44** includes rear brake discs **50** and rear brake calipers **52** operably coupled to rear wheel assemblies **12***a*.

(29) As shown in FIGS. **2-4**, braking assembly **40** also includes brake member **54**, illustratively a brake pedal, positioned within operator area **18** and is defined as one of the operator controls **24** (FIG. **1**). Brake member **54** is operably coupled to a brake master cylinder **56** such that braking input from the operator of vehicle **2** is applied to brake member **54** and is transmitted to brake master cylinder **56**.

(30) Referring still to FIGS. **2-4**, brake master cylinder **56** is operably coupled to a braking control system **58** which includes an anti-lock braking ("ABS") control module **60**. More particularly, brake master cylinder **56** is fluidly coupled to ABS control module **60** through conduit(s) or line(s) **62**. Illustratively, ABS control module **60** may be hydraulically actuated such that pressurized hydraulic fluid is configured to assist with the operation of braking assembly **40**. With the use of ABS control module **60**, braking assembly **40** is configured to operate in a normal run mode, in which an anti-lock braking feature ("ABS feature") is not engaged, and an anti-lock braking mode, in which the ABS feature is engaged.

(31) ABS control module **60** also is fluidly coupled with brake calipers **48**, **52**. Illustratively, as shown in FIGS. **2-4**, braking assembly **40** further includes a front left conduit or line **64**, a front right conduit or line **66**, a rear left conduit or line **68**, and a rear right conduit or line **70** which are all fluidly coupled to ABS control module **60** through four channels, namely a front left channel **140**, a front right channel **142**, a rear left channel **144**, and a rear right channel **146**, respectively (FIG. **10**). In this way, front left conduit **64** fluidly couples front left brake caliper **48***a* with ABS control module **60**, front right conduit **66** fluidly couples front right brake caliper **48***b* with ABS control module **60**, rear left conduit **68** fluidly couples rear left brake caliper **52***a* with ABS control module **60**, and rear right conduit **70** fluidly couples rear right brake caliper **52***b* with ABS control module **60**. ABS control module **60** also may include a front master cylinder output **148** and a rear master cylinder output **149**, both of which are operably coupled to brake master cylinder **56** (FIG. **10**), as disclosed herein.

(32) Referring to FIGS. **2-5**, with respect to rear end braking portion **44**, conduits **68**, **70** are fluidly coupled to ABS control module **60** through a junction member or box **72**. Illustratively, at least one junction conduit or line **74** (illustratively first and second junction conduits **74***a*, **74***b*) extends from ABS control module **60** to junction member **72** such that ABS control module **60** is fluidly coupled with rear brake calipers **52***a*, **52***b* through junction conduit **74**, junction member **72**, and respective rear left and right conduits **68**, **70**.

(33) As shown best in FIG. **5**, junction member **72** includes a first input **76** fluidly coupled to rear left conduit **68** through first junction conduit **74***a* and a second input **78** fluidly coupled to rear right conduit **70** through second junction conduit **74***b*. Junction member **72** facilitates serviceability of braking assembly **40** because if a repair or replacement is needed to rear end braking portion **44**, then the repair or replacement may be made at the location of junction member **72**, rather than having to fully disassemble all of braking assembly **40** for a repair to only a portion thereof. Additionally, junction member **72** is provided to allow for different braking pressures to be

transmitted to rear brake calipers **52***a*, **52***b*. For example, a first braking pressure may be provided to rear brake caliper **52***a* through first junction conduit **74***a* and rear left conduit **68** while a greater or lesser braking pressure may be provided rear brake caliper **52***b* through second junction conduit **74***b* and rear right conduit **70**.

(34) Referring now to FIG. **6**, braking control system **58** further includes front wheel speed sensors **80** configured to determine the rotational speed of front ground-engaging members **10** (FIG. **1**). Illustratively, each of front ground-engaging members **10** includes an individual wheel speed sensor **80**. In one embodiment, wheel speed sensor **80** is coupled to a portion of front final drive member **32** through fasteners **82**. As shown in FIG. **6**, wheel speed sensor **80** is received through an aperture **84** of a mounting bracket **86**. Mounting bracket **86** is coupled to a lateral portion of front final drive member **32** with fasteners **82** which are received within mounting bores **89** on the lateral portions of front final drive member **32**. More particularly, fasteners **82** are received within openings **83** on bracket **86**, which have an oval or oblong shape, thereby allowing the position of bracket **86** and sensor **80** to be adjustable relative to axle **37**. Additional fasteners or couplers **88** are configured to removably couple sensor **80** on mounting bracket **86**. It may be appreciated that sensor **80** is generally surrounded by mounting bracket **86** such that mounting bracket **86** conceals at least a portion of sensor **80** from debris and/or objects that may travel towards sensor **80** when vehicle **2** is moving, thereby minimizing damage to sensor **80** during operation of vehicle **2**.

(35) As shown best in FIG. **4**, each of front half shafts **37** includes a drive coupling with a splined shaft **106**. Splined shaft **106** couples with an output **112** (FIG. **6**) of front final drive member **32**. Additionally, a gear ring **108** is positioned on the outer surface of each of the drive couplings and is held in position relative to half shafts **37**. As such, gear ring **108** is configured to rotate with its corresponding half shaft **37**. Each of gear rings **108** includes a plurality of teeth **110** which cooperate with sensor **80** to determine the speed of each half shaft **37**. Sensors **80** are positioned in proximity to teeth **110** but do not contact teeth **110**; rather sensors **80** count teeth **110** as teeth **110** pass sensor **80** over a specific time period to calculate an angular velocity. Sensors **80** may be speed sensors such as Hall Effect speed sensors.

(36) Referring to FIG. **7**, braking control system **58** also includes rear wheel speed sensors **90** configured to determine the rotational speed of rear ground-engaging members **12** (FIG. **1**). Illustratively, each of rear ground-engaging members **12** includes an individual wheel speed sensor **90**. In one embodiment, wheel speed sensor **90** is coupled to a portion of rear final drive member **34**. As shown in FIG. **7**, wheel speed sensor **90** is received through an aperture **92** of a first mounting bracket **94** and is coupled to first mounting bracket **94** with fasteners **95**. It may be appreciated that sensor **90** is generally surrounded by first mounting bracket **94** such that mounting bracket **94** conceals at least a portion of sensor **90** from debris and/or objects that may travel towards sensor **90** when vehicle **2** is moving, thereby minimizing damage to sensor **90** during operation of vehicle **2**.

(37) First mounting bracket **94** is coupled to a second mounting bracket **96** through fasteners **98**. More particularly, fasteners **98** are received within openings **97** on first mounting bracket **94**, which have an oval or oblong shape, thereby allowing the position of first mounting bracket **94** and sensor **90** to be adjustable relative to axle **38**. And, second mounting bracket **96** is coupled to retainer members **100** on lateral portions of rear final drive member **34**. Additional fasteners or couplers **102** are configured to removably couple second mounting bracket **96** to retainers **100** because fasteners **102** are received through apertures **104** of retainers **100**. It may be appreciated that retainers **100** include a plurality of apertures **104** such that fasteners **102** can be received through any of apertures **104** to adjust the position of second mounting bracket **96** relative to axle **38**, thereby also allowing for the position of sensor **90** to be adjustable relative to axle **38**.

(38) As shown best in FIGS. **2** and **3**, each of rear half shafts **38** includes a drive coupling with a splined shaft **114** (FIG. **3**). Splined shaft **114** couples with an output (not shown) of rear final drive member **34**. Additionally, a gear ring **116** is positioned on the outer surface of each of the rear drive

couplings and is held in position relative to its corresponding rear half shaft **38**. As such, gear ring **116** is configured to rotate with its corresponding rear half shaft **38**. Each of gear rings **116** includes a plurality of teeth **118** which cooperate with sensor **90** to determine the speed of each rear half shaft **38**. Sensors **90** are positioned in proximity to teeth **118** but do not contact teeth **118**; rather sensors **90** count teeth **118** as teeth **118** pass sensor **90** over a specific time period to calculate an angular velocity. Sensors **90** may be speed sensors such as Hall Effect speed sensors.

(39) Referring to FIG. **8**, braking control system **58**, including ABS control module **60**, is electronically coupled or integrated with an overall electrical system **120** of vehicle **2**. Electrical system **120** of vehicle **2** includes an engine control module ("ECM") **122** and at least one display or gauge **124**. Display **124** is supported within operator area **18** (FIG. **1**) and is configured to provide information about vehicle **2** to the operator. In one embodiment, ABS control module **60** may be operated through display **124** such the operator may provide a user input or user selection through display **124**, which is transmitted to ABS control module **60**, to turn on/engage or turn off/disengage the ABS feature of braking assembly **40**. Illustrative display **124** may include toggle switches, buttons, a touchscreen, or any other type of surface or member configured to receive and transmit a selection made by the user. While ABS control module **60** is configured to engage/disengage the ABS feature through display **124** in the illustrative embodiment, it may be appreciated that vehicle **2** may include other inputs or means for engaging/disengaging the ABS feature.

(40) Additionally, ABS control module **60** is configured to transmit information about braking assembly **40** to display **124** to provide such information to the operator. For example, ABS control module **60** may be configured to transmit a fault signal to display **124** to indicate to the operator that a fault has occurred within a portion of braking assembly **40**, such as a fault of the ABS feature of braking assembly **40**. The fault indicator provided on display **124** may be a light, an alphanumeric code or message, or any other indication configured to alert the user of the fault.

(41) Additionally, display **124** is in electronic communication with ECM **122** to provide information to the operator about the engine (not shown) or other components of powertrain assembly **30**. Illustratively, ECM **122** transmits various signals to display **124** to provide information such as engine speed, engine temperature, oil pressure, the driving gear or mode, and/or any other information about powertrain assembly **30**. Additionally, as shown in FIG. **8**, display **124** is configured to provide inputs and other information to ECM **122**. For example, if illustrative vehicle **2** is configured with an adjustable speed limiting device and feature, the user may input speed limits to display **124** which are transmitted to ECM **122** from display **124** to control the speed of vehicle **2**, as disclosed further herein.

(42) Referring to FIG. **9**, a schematic view of braking control system **58** and at least a portion of electrical system **120** is disclosed with respect to operation of braking assembly **40**. As denoted, front end portion **14** and rear end portion **16** are shown and the left side of vehicle **2** is denoted with "L" and the right side of vehicle **2** is denoted with "R." As shown in FIG. **9**, when the operator depresses brake member **54** with a force F, force F is transmitted to brake master cylinder **56**, which may be a tandem master cylinder in one embodiment. Brake master cylinder **56** is configured to transmit braking input information to a brake pressure switch **126**. Brake pressure switch **126** is then configured to transmit a signal indicative of braking pressure information to a multi-pin connector **128**. Multi-pin connector **128** also may be configured to transmit and/or receive information to and from ECM **122**, a steering angle sensor **130** of electrical system **120**, display **124**, and ABS control module **60**. More particularly, ABS control module **60** may include a multi-axis G sensor **132** and a pressure sensor **134**, one or both of which may be internal or external sensors and are configured for communication with multi-pin connector **128**. Additionally, multi-pin connector **128** is electrically coupled with front wheel speed sensors **80** and rear wheel speed sensors **90**.

(43) Referring still to FIG. **9**, in operation, multi-pin connector **128** is configured to receive a user

input or user selection from display **124** to indicate if the user has turned on/engaged or turned off/disengaged the ABS feature of braking assembly **40** (e.g., via CAN messages) such that braking assembly **40** is to operate in the anti-lock braking mode or the normal run mode, respectively. Multi-pin connector **128** also may receive signals or other information from ECM **122**, steering angle sensor **130**, speed sensors **80**, **90**, multi-axis G sensor **132**, and pressure sensor **134** to determine information about the operating conditions of vehicle **2**. If the user has engaged the ABS feature of braking assembly **40**, for example through display **124**, such that braking assembly **40** is to operate in the anti-lock braking mode, then multi-pin connector **128** is configured to electrically communicate with ABS control module **60** to engage the ABS feature of braking assembly **40** when the user provides an input to brake member **54**.

(44) However, if the user has turned off/disengaged the ABS feature of braking assembly **40**, for example through a selection on display **124**, such that braking assembly **40** is to operate in the normal run mode, then multi-pin connector **128** is configured to determine if the ABS feature should be automatically turned on/engaged based on the vehicle operating conditions. For example, the ABS feature of braking assembly **40** may be automatically turned on/engaged based on a predetermined condition, such a vehicle operating condition, an environmental condition, or any other condition which may affect the driving conditions of vehicle **2**. In one embodiment, the predetermined condition may be a predetermined vehicle speed, the steering angle, conditions of the engine, terrain or environmental conditions, or any other condition or factor related to operating conditions of vehicle **2**. The predetermined vehicle speed that initiates automatic engagement of the ABS feature of braking assembly **40** may be approximately 30 kph. In this way, even if the user has previously selected to disengage the ABS feature of braking assembly **40**, the ABS feature will be automatically engaged, without any user input, via electrical system **120** (e.g., the communication between multi-pin connector **128** and ABS control module **60**) when vehicle **2** is operating at the predetermined operating condition, such as a vehicle speed of at least approximately 30 kph.

(45) It may be appreciated that the ABS feature of braking assembly **40** does not automatically turn off or disengage, but rather, is only disengaged through an operator or user input to display **124**. As such, the ABS feature may automatically engage based on vehicle operating conditions but does not automatically disengage and, instead, must be manually disengaged by the operator through display **124**. However, the ABS feature can only be disengaged when the vehicle speed is less than the predetermined vehicle speed (e.g., 30 kph). As such, even if the user selects, via display **124**, to disengage the ABS feature, ABS control module **60** will not disengage the ABS feature if the vehicle speed is greater than the predetermined vehicle speed. In one embodiment, display **124** may temporarily hide or conceal the user option to disengage the ABS feature when the vehicle speed is greater than the predetermined speed value (e.g., 30 kph).

(46) Also, even if a fault occurs in braking assembly **40**, the ABS feature will not disengage and, instead, a fault indicator will be provided to the operator through display **124**. Therefore, the operator will become aware of the fault within braking assembly **40** and may determine, based on the fault indication, if adjustments should be made to the operating conditions of vehicle **2**. Yet, the ABS feature will remain engaged throughout the fault condition. Furthermore, the fault indicator will not initiate a slow-down of the vehicle speed such that vehicle **2** may continue to operate at the speed input by the operator even when a fault is indicated.

(47) Referring now to FIG. **10**A, a schematic view of a hydraulic system **150** of vehicle **2** is disclosed with respect to operation of braking assembly **40**. Hydraulic system **150** includes a hydraulic reservoir **152** fluidly coupled to ABS control module **60** and also fluidly coupled to junction member **72**, and ground-engaging members **10**, **12** through any of conduits **64**, **66**, **68**, **70**, **74**. In operation, as force F is applied to brake member **54** by the operator, brake master cylinder **56** transmits force F to ABS control module **60** through at least brake pressure switch **126**. More particularly, brake master cylinder **56** is in communication with front and rear master cylinder outputs **148**, **149** which allows hydraulic fluid from hydraulic fluid reservoir **152** to flow to front

and rear ground-engaging members **10**, **12** through channels **140**, **142**, **144**, **146**.

(48) Illustratively, and still referring to FIG. **10**A, as force F is applied to brake member **54**, brake master cylinder **56** provides an input to front master cylinder output **148** through brake pressure switch **126** to initiate a flow of hydraulic fluid through front left channel **140** and front left conduit **64** to front left ground-engaging member **10**. Additionally, the input provided to front master cylinder output **148** through brake pressure switch **126** also initiates a flow of hydraulic fluid through front right channel **142** and front right conduit **66** to front right ground-engaging member **10**. With respect to rear ground-engaging members **12**, as force F is applied to brake member **54**, brake master cylinder **56** provides an input to rear master cylinder output **149** to initiate a flow of hydraulic fluid through rear left channel **144**, first junction conduit **74***a*, junction member **72**, and rear left conduit **68** to rear left ground-engaging member **12**. Additionally, the input provided to rear master cylinder output **149** from brake master cylinder **56** also initiates a flow of hydraulic fluid through rear right channel **146**, second junction conduit **74***b*, junction member **72**, and rear right conduit **70** to rear right ground-engaging member **12**. In this way, a single actuation of braking assembly **40** when the operator depresses brake member **54** allows for braking of all ground-engaging members **10**, **12** through the four channels **140**, **142**, **144**, **146** of ABS control module **60**. It may be appreciated that, if the ABS feature is engaged, the flow of hydraulic fluid to any of brake calipers **48**, **52** may be modulated, temporarily stopped, and/or otherwise adjusted by ABS control module **60** to minimize skidding and maintain steering control of vehicle **2**.

(49) Referring now to FIG. **10**B, a schematic view of an alternative hydraulic system **150**′ of vehicle **2** is disclosed with respect to operation of braking assembly **40**, with like components of hydraulic system **150** (FIG. **10**A) shown with like reference numbers. In operation, as force F is applied to brake member **54** by the operator, brake master cylinder **56** transmits force F to ABS control module **60**. More particularly, as force F is applied to brake member **54**, brake master cylinder **56** provides an input to front master cylinder output **148** to initiate a flow of hydraulic fluid through front left channel **140** and front left conduit **64** to front left ground-engaging member **10**. Additionally, the input provided to front master cylinder output **148** also initiates a flow of hydraulic fluid through front right channel **142** and a first front right conduit **164** fluidly coupled to a junction block or junction member **162**. First front right conduit **164** is fluidly coupled to a first switching member **126**′ of junction member **162** and transmits hydraulic fluid or other braking input or signal to front right ground-engaging member **10** through second front right conduit **66**.

(50) With respect to rear ground-engaging members **12**, as force F is applied to brake member **54**, brake master cylinder **56** provides an input to rear master cylinder output **149** to initiate a flow of hydraulic fluid through rear right channel **146**, junction conduit **74**, junction member **72**, and rear right conduit **70** to rear right ground-engaging member **12**. Additionally, the input provided to rear master cylinder output **149** from brake master cylinder **56** also initiates a flow of hydraulic fluid through rear left channel **144** which is fluidly coupled to junction member **162** through a first junction conduit **168**. At junction member **162**, hydraulic fluid or other braking input or signal is transmitted through a second switching member **126**″ and flows through second junction conduit **166**, which is fluidly coupled to junction member **72**. At junction member **72**, hydraulic fluid or other braking input flows through rear left conduit **68** to rear right ground-engaging member **12**. In this way, a single actuation of braking assembly **40** when the operator depresses brake member **54** allows for braking of all ground-engaging members **10**, **12** through the four channels **140**, **142**, **144**, **146** of ABS control module **60**. It may be appreciated that, if the ABS feature is engaged, the flow of hydraulic fluid to any of brake calipers **48**, **52** may be modulated, temporarily stopped, and/or otherwise adjusted by ABS control module **60** to minimize skidding and maintain steering control of vehicle **2**.

(51) ABS Operating Modes

(52) With respect to the operation of braking assembly **40**, FIGS. **11-13** disclose various operating modes which may be used for the ABS feature. As disclosed herein, braking assembly **40** may be

configured to automatically turn off the ABS feature when the vehicle speed is below a prescribed or predetermined speed (e.g., 30 kph) and is configured to automatically turn on the ABS feature when the vehicle speed is above the prescribed speed. Alternatively, braking assembly **40** may be configured to allow the user to manually turn on and off the ABS feature.

(53) FIG. **11** discloses a first operating mode of braking assembly **40** in which the ABS feature always engages when braking is applied by the user (i.e, the "ABS On Mode"). More particularly, if vehicle **2** first begins to operate in the normal run mode (i.e., the ABS feature is not initially engaged), electrical system **120** may determine if brake member **54** (FIG. **2**) has been actuated such that braking has been applied in a Step **200**. If no braking has been applied, then vehicle **2** continues to operate in the normal run mode, as controlled by the operator, in Step **202**. However, if braking is applied in Step **200**, for example through brake member **54**, then ABS control module **60** takes control of braking assembly **40** in Step **204** and, because the ABS feature is always engaged in this ABS On Mode, the ABS feature is utilized in the braking process. In Step **204**, ABS control module **60** receives inputs from at least pressure sensor **134**, brake pressure switch **126**, wheel speed sensors **80**, **90**, ECM **122**, and display **124**. Display **124** also may receive inputs from a portion of powertrain assembly **30**, such as the engine and/or transmission (not shown), regarding information of the operating conditions thereof. With this information, ABS control module **60** modulates cycles of brake pressure, using hydraulic fluid from hydraulic fluid reservoir **152** (FIG. **10**A), to distribute pressurized braking fluid to at least some of ground-engaging members **10**, **12** in Step **206**. During Step **206**, ABS control module **60** modulates the pressured braking fluid based on information received by wheel speed sensors **80**, **90** to obtain appropriate vehicle deceleration through different wheel slips. Once vehicle **2** properly decelerates and braking has been terminated, vehicle **2** returns to the normal run mode in Step **208** until another braking input is applied, at which time, the ABS feature will automatically be utilized again by ABS control module **60** in the ABS On Mode.

(54) However, and now referring to FIG. **12**, as disclosed herein, the ABS feature of braking assembly **40** may be selectively engaged in a second operating mode (i.e., the "ABS On-Off Mode"). More particularly, in the ABS On-Off Mode, the ABS feature may not always be engaged upon a braking input, but instead, the operator may selectively engage or disengage the ABS feature through display **124**. As such, FIG. **12** discloses a Step **210** which allows electrical system **120** to determine if a user input or user selection has been applied to display **124** to engage or disengage the ABS feature such that vehicle **2**. If the vehicle **2** is operating in the normal run mode and no input has been provided to display **124**, then vehicle **2** continues in the normal run mode and is controlled by the operator, as shown in Step **212**.

(55) However, if display **124** has disengaged the ABS feature of braking assembly **40** in Step **210**, then electrical system **120**, including ABS control module **60**, determines if the vehicle speed is below a predetermined value (e.g., 30 kph) in Step **214** using information from wheel speed sensors **80**, **90** (FIG. **9**). If the vehicle speed is less than the predetermined speed threshold (e.g., 30 kph), then ABS control module **60** determines if braking is sensed by an input applied to brake member **54** (FIG. **2**) in Step **216**. If braking has not been sensed, then vehicle **2** continues to operate in the normal run mode and is controlled by the operator, as shown in Step **218**. And, even if braking has been sensed, as long as the vehicle speed is below the predetermined operating condition (e.g., vehicle speed of 30 kph), then ABS control module **60** allows braking to occur without the ABS feature, as shown in Step **220**. In Step **222**, vehicle **2** will achieve braking deceleration without the ABS feature engaged and will return to the normal run mode. In this way, the ABS feature of braking assembly **40** will not automatically engage if the user has disengaged the ABS feature and the vehicle speed is below the predetermined threshold value. As such, braking may occur but without the ABS feature engaged when braking assembly **40** operates in the ABS On-Off Mode.

(56) Yet, as shown in Step **224** of FIG. **12**, if the vehicle speed is above the predetermined

threshold value (e.g., 30 kph), then ABS control module **60** is automatically engaged to take control of operation of braking assembly **40** and engage the ABS feature, despite the user's previous selection through display **124** to disengage the ABS feature. In this way, ABS control module **60** will automatically change from the normal run mode to the anti-lock braking mode in response to the vehicle speed, regardless of the user's previous selection with respect to the ABS feature. At Step **224**, ABS control module **60** receives inputs, signals, or other information from a plurality of other components, such as brake pressure switch **126**, pressure sensor **134**, wheel speed sensors **80**, **90**, ECM **122**, and display **124**. Using this information, ABS control module **60** then modulates cycles of brake pressure, using hydraulic fluid from hydraulic fluid reservoir **152** (FIG. **10**A), to distribute pressurized braking fluid to each of ground-engaging members **10**, **12** in Step **226**. During Step **226**, ABS control module **60** (e.g., internal solenoids) modulates the pressured braking fluid based on information received by wheel speed sensors **80**, **90** to obtain appropriate vehicle deceleration through different wheel slips. Once vehicle **2** properly decelerates, vehicle **2** returns to the normal run mode in Step **228** until another braking input is applied.

(57) Referring to FIG. **13**, a third operating mode of braking assembly **40** is shown as the ABS Control Module Mode. More particularly, as the operator provides an input to brake member **54** (FIG. **2**), the brake input is transmitted to brake master cylinder **56** to start the braking process, as shown in Step **230**. The braking pressure may be determined by brake pressure switch **126** and pressure sensor **134** (FIG. **10**A) in Step **232** and the speed of ground-engaging members **10**, **12** is determined by respective wheel speed sensors **80**, **90**, as shown in Step **234**.

(58) With this information from Steps **232** and **234**, electrical system **120**, including ABS control module **60**, may determine if the rate of deceleration of any of ground-engaging members **10**, **12** is greater than the deceleration rate of the other ground-engaging members **10**, **12**, as shown in Step **236**. If the deceleration rate of one of ground-engaging members **10**, **12** is not greater than that of the others, then brake pressure is maintained until brake member **54** is released by the operator, as shown in Step **238**.

(59) However, if the deceleration rate of one of ground-engaging members **10**, **12** is greater than that of the others, then, in Step **240**, ABS control module **60** (e.g., internal solenoid) may release the brake pressure on the one ground-engaging member **10**, **12** which has a greater deceleration rate than the others. In Step **240**, ABS control module **60** is configured to release the brake pressure on the one ground-engaging member **10**, **12** until the remaining ground-engaging members **10**, **12** increase in deceleration rate to equal that of the one ground-engaging member **10**, **12**. In this way, ABS control module **60** utilizes the ABS feature to minimize wheel slipping on the ground surface and to maintain steering control of vehicle **2**.

(60) Once all ground-engaging members **10**, **12** have approximately equal deceleration rates, then ABS control module **60** re-applies braking pressure to the one ground-engaging member **10**, **12** with the initially-greater deceleration rate such that braking pressure is now applied to all of ground-engaging members **10**, **12**, as shown in Step **242**.

(61) It may be appreciated that braking assembly **40** may be pre-set to operate in only one of the three operating modes of FIGS. **11-13**, as set by a manufacturer or dealer of vehicle **2**, or may be configured to operate in any of the operating modes of FIGS. **11-13** based on an input from the user. It may be appreciated that, in any of the three operating modes of FIGS. **11-13**, ABS control module **60** may be automatically engaged to turn on the ABS feature in response to an error of the vehicle speed transmitted by ECM **122** and/or sensors **80**, **90**. Additionally, depending on the operating mode, the user has the ability to turn on and off the ABS feature during operation of vehicle **2** and, as such, can make adjustments to the performance and handling of vehicle **2** while operating vehicle **2**. However, as disclosed herein, electrical system **120** may ignore a user's request to disengage or turn off the ABS feature, depending on the predetermined vehicle condition (e.g., a vehicle speed of at least 30 kph).

(62) Furthermore, it may be appreciated that both the ABS On Mode and the ABS On-Off Mode

may utilize the features of the ABS Control Mode by also modulating the braking pressure, as disclosed best in FIG. **13**, in Steps **206** and **226**, respectively. As such, when the ABS feature is engaged, ABS control module **60** is configured to monitor the deceleration rate of each of ground-engaging members **10**, **12** and may regulate or modulate the flow of hydraulic flow to any brake caliper **48**, **52** of a ground-engaging member **10**, **12** with a greater deceleration rate than the others.
(63) ASLD Operating Modes
(64) Referring to FIG. **14**, vehicle **2** may be configured with an adjustable speed limiting device or feature ("ASLD") in which the user may selectively define speed limits when vehicle **2** is operating. For example, the user may engage or turn on the adjustable speed limiting feature through display **124**, which will allow vehicle **2** to operate at speed limits between a predetermined lower speed limit (e.g., 30 kph) and a predetermined maximum speed limit. When utilizing the adjustable speed limiting feature, the user may adjust the speed limit in predetermined speed intervals or increments (e.g., 5 kph) for each step change.
(65) As shown in FIG. **14**, in Step **170**, vehicle **2** operates in the normal run mode. While vehicle **2** operates in the normal run mode, display **124** is in a corresponding normal run mode, as shown in Step **172**. In Step **174**, the operator or another user enters Display Menu Options by pressing or otherwise providing an input to a "MODE" input of display **124**. In Step **176**, the user scrolls through the Display Menu Options with inputs, such as "UP" and "DOWN" arrow buttons. In Step **178**, the user may select and enter the ASLD Display Menu Options. If the user does not select the ASLD Display Menu Options, then the user may select an "EXIT" input in Step **179**. If the user selects the "EXIT" option in Step **179**, then display **124** returns to the normal run mode, as shown in Step **172**. However, if the user does not select the "EXIT" option in Step **179**, then the user is able to continue to scroll through the Display Menu Options, as shown in Step **176**.
(66) If, in Step **178**, the user selects and enters the ASLD Display Menu Options, then vehicle **2** begins to operate in accordance with the ASLD feature in Step **180**. When utilizing the ASLD feature in Step **180**, electrical system **120**, such as ABS control module **60**, may communicate with display **124**, ECM **122**, and wheel speed sensors **80**, **90** to obtain any necessary information for operating vehicle **2** according to the ASLD feature. In Step **182**, display **124** provides or shows the actual vehicle speed as well as a user selectable speed limit. The user selectable speed may be labeled on display as "Set Speed", "Speed Lim", or any other type of alphanumeric code, label, or information that alerts the user to the location of the user selectable speed limit option on display **124**. The user selectable speed limit may be initialized to the maximum speed of vehicle **2**, rounded to the nearest 5 kph, in one embodiment.
(67) In Step **184**, the user selectable speed limit variable may be updated to any value selected by the user and stored in ECM **122**. In Step **186**, ECM **122** provides the updated user selected speed limit to display **124** such that the user is able to quickly determine the speed limit. In Step **188**, ECM **122** will not allow vehicle **2** to travel faster than the user selected speed limit.
(68) In Step **190**, an input (e.g., a button) may be actuated (e.g., pressed) on display **124** when the user is operating vehicle **2** in accordance with the ASLD feature. For example, if the "MODE" input is actuated in Step **190**, then vehicle **2** continues to operate in accordance with the ASLD feature.
(69) However, if the "DOWN" input (e.g., arrow button) is actuated in Step **190**, then display **124** sends a decrement command to ECM **122** via the CAN bus network for a possible reduction to the user selected speed limit, as shown in Step **191**. In Step **192**, it is determined if the current user selected speed limit is greater than a predetermined speed value (e.g., 30 kph). If Step **192** determines that the user selected speed limit is greater than the predetermined speed value, then ECM **122** decreases the value of the user selected speed limit by a predetermined incremental amount (e.g., 5 kph), as shown in Step **193**. Following Step **193**, ECM **122** is updated with the decreased user selected speed limit, as shown in Step **184**.
(70) Yet, if Step **192** determines that the user selected speed limit is not greater than the

predetermined speed value, then, as shown in Step **194**, the request to modify the user selected speed limit through display **124** is ignored by ECM **122** and the original user selected speed limit continues to be stored in ECM **122**, as shown in Step **184**.

(71) However, if in Step **190**, if the “UP” input (e.g., arrow button) is actuated, then, as shown in Step **195**, display **124** sends an increment command to ECM **122** via the CAN bus network for a possible increase to the user selected speed limit. In Step **196**, it is determined if the current user selected speed is less than a calibrated maximum vehicle speed for vehicle **2**. If Step **195** determines that the user selected speed limit is less than the calibrated maximum vehicle speed, then ECM **122** increase the value of the user selected speed limit by a predetermined incremental amount (e.g., 5 kph), as shown in Step **197**. Following Step **197**, ECM **122** is updated with the increased user selected speed limit, as shown in Step **184**.

(72) Yet, if Step **196** determines that the user selected speed limit is greater than the calibrated maximum vehicle speed, then the request to modify the user selected speed limit through display **124** is ignored by ECM **122**, as shown in Step **194**.

(73) Additionally, in one embodiment, if vehicle **2** is operating in various modes (e.g., a farm or ranch mode), the user may first shift to low gear before engaging the adjustable speed limiting feature through display **124**. Once the adjustable speed limiting feature is engaged, the predetermined speed increments for each step change may be approximately 1 mph. For example, in an embodiment of vehicle **2** having the farm or ranch operating mode, the predetermined lower speed limit may be approximately 5 mph and the predetermined maximum speed limit may be approximately 12 mph with predetermined speed increments of approximately 1 mph for each step change.

(74) ESC Operating Modes

(75) Additionally, as shown in at least FIG. **9**, with the addition of steering angle sensor **130**, ECM **122**, ABS control module **60**, and/or any other component of electrical system **120** may include an electronic stability control (“ESC”) assembly or program **160**. ESC assembly **160** may include a yaw rate sensor positioned within a portion of steering assembly **26** (FIG. **1**), for example in a portion of an electric power steering module, as well as steering angle sensor **130**. ESC assembly **160** may be configured within ECM **122**, any other component of electrical system **120**, and/or may be a separate module electrically coupled to electrical system **120** and/or ECM **122**. In one embodiment, ESC assembly **160** may be selectively engaged by the user through display **124** and/or any other component of vehicle **2**; however, in other embodiments, ESC assembly **160** may be automatically engaged by ECM **122** or other components of electrical system **120** based on various operating conditions, such as vehicle conditions, environmental conditions, terrain conditions, etc. Also, it may be appreciated the ESC assembly **160** may always be engaged upon starting vehicle **2** such that ESC assembly **160** is not selectively engaged or disengaged.

(76) More particularly, and as shown in FIGS. **15-20**, ESC assembly **160** is configured to operate in various operating modes. With respect to FIG. **15**, ESC assembly **160** is configured to operate in a first or ESC and ABS Normal Operating Mode. In the ESC and ABS Normal Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **250**, electrical system **120** may determine if brake member **54** (FIG. **2**) has been actuated such that braking has been applied in a Step **252**. If no braking has been applied, then vehicle **2** continues to operate in the normal run mode, as controlled by the operator, in Step **254**.

(77) However, if braking is applied in Step **252**, for example through brake member **54**, then ABS control module **60** takes control of braking assembly **40** in Step **256**. In Step **256**, ABS control module **60** may actively request prescribed drag torque reduction to ECM **122**. Additionally, in Step **256**, ABS control module **60** communicates with at least pressure sensor **134**, brake pressure switch **126**, wheel speed sensors **80**, **90**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine and/or transmission (not shown), regarding information of the operating conditions thereof, and ECM **122** may

communicate with other components of vehicle **2**.

(78) With this information, ABS control module **60** modulates cycles of brake pressure, using hydraulic fluid from hydraulic fluid reservoir **152** (FIG. **10**A), to distribute pressurized braking fluid to each brake caliper **48**, **52** in Step **258**. During Step **258**, using information from speed sensors **80**, **90**, ABS control module **60** modulates the pressured braking fluid based on information received by wheel speed sensors **80**, **90** to obtain appropriate vehicle deceleration through different wheel slips. Once vehicle **2** properly decelerates and braking has been terminated, vehicle **2** returns to the normal run mode in Step **260** until another braking input is applied.

(79) With respect to FIG. **16**, ESC assembly **160** is configured to operate in a second or Hill Descent Control ("HDC") Operating Mode. In the HDC Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **262**, electrical system **120** may determine if accelerator member **53** (FIG. **2**) has been released such that at least no acceleration is being applied in a Step **264**. If accelerator member **53** has not been released, then vehicle **2** continues to operate in the normal run mode, as controlled by the operator pressing or otherwise providing an input to accelerator member **53**, in Step **266**. In Step **266**, vehicle **2** continues to operate in the normal run mode while the operator provides an input to accelerator member **53** as long as the vehicle speed is less than a predetermined speed value (e.g., 4 mph).

(80) However, if accelerator member **52** has been released in Step **264**, then ABS control module **60** takes control of braking assembly **40** in Step **268**. In Step **268**, ABS control module **60** may monitor inputs from speed sensors **80**, **90** to apply an appropriate amount of brake pressure to each caliper **48**, **52** to decrease vehicle speed while maintaining a predetermined or specified vehicle speed deceleration rate and proper wheel slips. In one embodiment, the predetermined or specified vehicle speed deceleration rate may be approximately 4 mph when vehicle **2** is traveling down hill. Additionally, in Step **268**, ABS control module **60** communicates with at least pressure sensor **134**, brake pressure switch **126**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine and/or transmission (not shown), regarding information of the operating conditions thereof, and ECM **122** may communicate with other components of vehicle **2**. In one embodiment, in Step **268**, ECM **122** communicates with the engine to determine torque and rpm information and also may communicate with accelerator member **53** for electronic throttle control (FIG. **2**).

(81) In Step **270**, ESC assembly **160** is configured to maintain speed and/or a speed reduction to the prescribed speed vehicle speed (e.g., 4 mph), until vehicle **2** stops, or until the operator provides an input to accelerator member **53** (i.e., invokes a speed larger than the prescribed vehicle speed (e.g., 4 mph), thereby releasing the braking input).

(82) With respect to FIG. **17**, ESC assembly **160** is configured to operate in a third or Hill Assist/Hill Hold Control ("HHC") Operating Mode. In the HHC Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **272**, electrical system **120** may determine if vehicle **2** has stopped moving while in an uphill direction and if brake member **54** (FIG. **2**) has been sufficiently applied to retain vehicle **2** in a stationary position while on uphill terrain or in an uphill direction, as shown in Step. **274**. If it is determined that vehicle **2** has stopped moved in the uphill direction but the brake torque is insufficient to keep vehicle **2** from rolling backwards in a downhill direction, as shown in Step **276**, then the HDC Operating Mode is invoked, as shown in Step **278**, to prevent vehicle **2** from moving or rolling backwards in the downhill direction.

(83) If, however, in Step **274**, it is determined that vehicle **2** has stopped moving in the uphill direction but sufficient braking torque is provided to retain vehicle **2** in a stationary position while on the uphill terrain, then ABS control module **60** takes control of braking commands in Step **280**. Additionally, in Step **280**, ABS control module **60** may monitor inputs of braking pressure applied to brake member **54** by the operator and any sensed changed in the G-force, as sensed by multi-axis g sensor **132** (FIG. **9**), in the uphill direction. ABS control module **60** also may monitor any input to acceleration member **53** which may be provided to the engine to increase or change engine

torque and speed. Also, in Step **280**, ABS control module **60** communicates with at least pressure sensor **134**, brake pressure switch **126**, speed sensors **80**, **90**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine and/or transmission (not shown), regarding information of the operating conditions thereof, and ECM **122** may communicate with other components of vehicle **2**. In one embodiment, in Step **268**, ECM **122** communicates with the engine to determine torque and rpm information and also may communicate with accelerator member **53** (FIG. **2**) for electronic throttle control ("ETC").

(84) In Step **282**, ESC assembly **160** is configured to appropriately maintain static brake pressure, as applied by the operator, which is the same brake pressure sufficient to hold vehicle **2** in a stationary position for a predetermined amount of time (e.g., 1.0-5.0 seconds and, more particularly, 1.5-3.0 seconds). Alternatively, in Step **282**, ESC assembly **160** is configured to maintain static brake pressure until engine torque is applied by the operator, for example through accelerator member **53** (FIG. **2**), in an amount which overcomes the braking torque. ABS control module **60** may adjust the pressure upon input from ETC, engine torque, and/or engine speed information through the CAN network or messages.

(85) In Step **284**, vehicle **2** may return to the normal run mode in the uphill direction or may roll backwards in the downhill direction if vehicle **2** is maintained on the uphill terrain for a time greater than the predetermined amount of time (e.g., 1.5-3.0 seconds). If vehicle **2** returns to the normal run mode in Step **284**, then the HHC Operating Mode returns to Step **272**. However, if vehicle **2** begins to roll or move backwards in the downhill direction in Step **284**, then the HHC Operating Mode returns to Step **278** to prevent such movement.

(86) Referring to FIG. **18**, ESC assembly **160** is configured to operate in a fourth or Roll Over Mitigation ("ROM") Operating Mode. In the ROM Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **290**, electrical system **120** may determine if vehicle **2** is moving, turning to the left or right, and/or operating at an acceleration within the G-force specifications for vehicle **2**, as shown in Step **292**. If electrical system **120** determines that vehicle is not moving, turning to the left or right, and/or operating at an acceleration within the G-force specifications, then vehicle **2** continues to operate in a normal straight, left-turn, or right-turn driving mode, as shown in Step **294**.

(87) However, if Step **292** determines that vehicle **2** is moving, turning to the left or right, and/or operating at an acceleration within the G-force specifications for vehicle **2**, then Step **296** determines if the lateral acceleration of vehicle **2** is greater than the predetermined or set intervention ROM value. In other words, Step **296** determines if vehicle **2** is likely to tip over. If it is determined that vehicle **2** is likely to tip over, then, in Step **298**, ABS control module **60** takes control of braking commands and monitors the G forces, wheel speed, steering angle, and steering angle rate of change using sensors **132**, **80** and **90**, and **130**, respectively. Additionally, in Step **298**, ABS control module **60** may communicate with at least pressure sensor **134**, brake pressure switch **126**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine and/or transmission (not shown), regarding information of the operating conditions thereof, and ECM **122** may communicate with other components of vehicle **2**. In one embodiment, in Step **298**, ECM **122** communicates with the engine to determine torque and rpm information and also may communicate with accelerator member **53** (FIG. **2**) for electronic throttle control ("ETC").

(88) In Step **300**, ABS control module **60** is configured to appropriately administer brake pressure to each of calipers **48**, **52** with appropriate amounts of wheel slips to obtain a normal lateral stability value (i.e., a stability value within a predetermined range). In Step **302**, vehicle **2** returns to the normal run and steering mode.

(89) Referring to FIG. **19**, ESC assembly **160** is configured to operate in a fifth or Traction Control System ("TCS") Operating Mode. In the TCS Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **310**, electrical system **120** may determine if the operator is

applying an input to accelerator member **53** (FIG. **2**), thereby causing engine torque and speed to cause wheel slips, as shown in Step **312**. If Step **312** determines that the operator is not applying an input to accelerator member **53** in a manner resulting in engine torque and speed causing wheel slips, then vehicle **2** continues to operate in the normal run mode and according to normal driving conditions, as shown in Step **314**.

(90) However, if Step **312** determines that the operator is applying an input to accelerator member **53** in a manner resulting in engine torque and speed causing wheel slips, then, in Step **316**, ABS control module **60** takes control of acceleration commands and actively communicates with ECM **122** and wheel speed sensors **80**, **90** to reduce engine torque and speed. To reduce engine torque and speed, Step **316** applies an amount of braking pressure to each of brake calipers **48**, **52** according to different drive modes, such as a Turf or 4×1 mode, a 4×2 mode, 4×4 mode, reverse, and any other type of mode configured for vehicle **2**. Additionally, in Step **316**, ABS control module **60** may communicate with at least pressure sensor **134**, brake pressure switch **126**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine, transmission, drive shafts, and wheel assemblies **10***a*, **12***a* (FIG. **2**), regarding information of the operating conditions thereof, and ECM **122** may communicate with other components of vehicle **2**.

(91) In Step **318**, ABS control module **60** appropriately modulates the braking pressure distributed to each of brake calipers **48**, **52** in combination with appropriate modulations of engine torque reductions. In Step **320**, vehicle **2** returns to the normal run mode and operates according to normal driving conditions.

(92) Referring to FIG. **20**, ESC assembly **160** is configured to operate in a sixth or Vehicle Dynamic Control (“VDC”) Operating Mode. In the VDC Operating Mode, when vehicle **2** is operating in the normal run mode, as shown in Step **322**, electrical system **120** may determine if vehicle **2** is moving, turning to the right or the left, pitching, and/or braking in a manner less than predetermined values for such operations, as shown in Step **324**. If Step **324** determines that vehicle **2** is not moving, turning to the right or the left, pitching, and/or braking in a manner less than predetermined values for such operations, then vehicle **2** operates in the normal run mode and/or according to normal driving conditions with regular steering and braking parameters, as shown in Step **326**.

(93) However, if Step **324** determines that vehicle **2** is moving, turning to the right or the left, pitching, and/or braking in a manner less than predetermined values for such operations, then, in Step **328**, ABS control module **60** takes control of braking by applying optimum wheel speed parameters using wheel speed sensors **80**, **90**. More particularly, ABS control module **60** requests engine torque reductions, monitors the steering angle and steering rate using sensor **130**, and monitors the changes in G-forces using sensor **132**. Additionally, in Step **328**, ABS control module **60** may communicate with at least pressure sensor **134**, brake pressure switch **126**, ECM **122**, and display **124**. Display **124** also may communicate with a portion of powertrain assembly **30**, such as the engine and/or transmission, regarding information of the operating conditions thereof, and ECM **122** may communicate with other components of vehicle **2**. In one embodiment, in Step **328**, ECM **122** communicates with the engine to determine torque and rpm information and also may communicate with accelerator member **53** (FIG. **2**) for electronic throttle control (“ETC”).

(94) In Step **330**, ABS control module **60** appropriately applies braking pressure and proper wheel slips to each of ground-engaging members **10**, **12** to maintain the intended direction (i.e., steering) and to maintain stability, thereby preventing oversteer or understeer. ABS control module **60** also monitors the lateral, longitudinal, pitch, and/or yaw directions. In Step **332**, vehicle returns to the normal run mode and/or normal driving, steering, and braking parameters.

(95) Additional details of braking assembly 40 may be disclosed in U.S. patent application Ser. No. 15/471,469, filed Mar. 28, 2017, and entitled “ANTI-LOCK BRAKE SYSTEM FOR ALL-TERRAIN VEHICLE”, the complete disclosure of which is expressly incorporated by reference

herein.
(96) While this invention has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. A vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one brake caliper, the at least one brake caliper coupled to a first ground engaging member of the plurality of ground engaging members and configured to control brake pressure to the first ground engaging member; a powertrain supported by the frame, the powertrain operably coupled to at least one ground engaging member of the plurality of ground engaging members; an accelerator input configured to receive an input from an operator, and operably coupled to the powertrain, the accelerator member configured to have a first state and a second state; at least one sensor supported by the plurality of ground engaging members; a controller operably coupled to the powertrain and the at least one sensor; and a brake control module operably coupled to the at least one brake caliper, the controller, and at least one sensor, and in response to the accelerator input being in the first state, the brake control module operates in a first operational mode, and in response to the accelerator input being in the second state, the brake control module operates in a second operational mode.
2. The vehicle of claim 1, wherein the first state is an actuated state and the first operational mode is a normal run mode.
3. The vehicle of claim 2, wherein the second state is a released state and the second operational mode is a hill descent control mode configured to maintain a predetermined vehicle speed.
4. The vehicle of claim 3, wherein, in the second operational mode, the at least one sensor is a speed sensor, and the brake control module is configured to monitor a vehicle speed from the speed sensor and control the brake pressure to maintain a prescribed speed.
5. The vehicle of claim 4, wherein the brake control module operates in the second operational mode until the accelerator input is changed to the first state.
6. The vehicle of claim 4, wherein the brake control module operates in the second operational mode until the vehicle stops.
7. The vehicle of claim 1, further comprising a display, and the display is operable to communicate information regarding operating conditions of the vehicle.
8. The vehicle of claim 1, further comprising a display, and the display is operably coupled to each of the controller and the brake control module.
9. A method of operating a vehicle with a plurality of ground engaging members and at least one brake member, the at least one brake member operably coupled to a first ground engaging member of the plurality of ground engaging members and an engine operably coupled to at least one ground engaging member of the plurality of ground engaging members, the method comprising: determining the vehicle has stopped moving while in an uphill direction; operating the vehicle according to a first operating condition in response to the vehicle having stopped moving in an uphill direction to retain the vehicle in a stationary position; determining the vehicle is moving backwards in a downhill direction; and operating the vehicle according to a second operating condition in response to the vehicle moving backwards in a downhill direction to prevent the vehicle from moving backwards in the downhill direction.
10. The method of claim 9, wherein the first operating condition includes operating a brake assembly to maintain brake pressure at the at least one brake member to retain the vehicle in a stationary position.

11. The method of claim 9, further comprising: determining an engine torque is requested by an operator; and operating the vehicle according to a third operating condition.
12. The method of claim 11, wherein the third operating condition is a normal run operating condition.
13. A vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one brake caliper, the at least one brake caliper coupled to a first ground engaging member of the plurality of ground engaging members and configured to control brake pressure to the first ground engaging member; a powertrain supported by the frame, the powertrain operably coupled to at least one ground engaging member of the plurality of ground engaging members; an accelerator input configured to receive an input from an operator, and operably coupled to the powertrain, the accelerator member configured to have a first state and a second state; at least one sensor supported by the plurality of ground engaging members, the sensor configured to determine an orientation of the vehicle; a controller operably coupled to the powertrain and the at least one sensor; and a brake control module operably coupled to the controller and the at least one brake caliper, and (a) at a first time in response to the vehicle being stopped in an uphill direction, the brake control module is configured to operate in a first operational mode to control the at least one brake caliper to retain the vehicle in a stationary position for a predetermined time period, and (b) at a second time subsequent to the first time, in response to the vehicle moving backwards in a downhill direction, the controller is configured to operate in a second operational mode to control the at least one brake caliper to control the vehicle moving in the downhill direction.
14. The vehicle of claim 13, wherein the predetermined time period is between one to five seconds.
15. The vehicle of claim 13, wherein the second operational mode includes controlling the vehicle to prevent the vehicle from moving.
16. The vehicle of claim 13, wherein the second operational mode includes controlling the vehicle to move backwards in a downhill direction at a constant speed.
17. The vehicle of claim 13, further comprising a display, and the display operable to communicate information regarding operating conditions of the vehicle.
18. The vehicle of claim 13, further comprising a display, and the display operably coupled to each of the controller and the brake control module.
19. A vehicle, comprising: a plurality of ground engaging members; a frame supported by the plurality of ground engaging members; at least one brake caliper, the at least one brake caliper coupled to a first ground engaging member of the plurality of ground engaging members; an electrical system comprising: a control module operably coupled to the at least one brake caliper; a first sensor configured to measure a first characteristic of the vehicle; and a second sensor configured to measure a second characteristic of the vehicle; and in response to the first characteristic of the vehicle exceeding a first threshold and the second characteristic of the vehicle exceeding a second threshold, operating the at least one brake caliper to induce a wheel slip in the first ground engaging member.
20. The vehicle of claim 19, wherein the first characteristic is an acceleration of the vehicle.
21. The vehicle of claim 20, wherein the acceleration value is a lateral acceleration.
22. The vehicle of claim 20, wherein the second characteristic is one of a steering characteristic and a wheel speed.
23. The vehicle of claim 19, wherein the electrical system further comprises a display, and the display is configured to communicate information related to the operating conditions of the vehicle.
24. The vehicle of claim 19, wherein the electrical system further comprises a display, and the display operably coupled to each of the controller and the brake control module.